

V3 Dashboard Enhancements

HRAlit Interactive Dashboard - April 2026

Seven UI and data improvements applied to the HRAlit Dashboard, with the most significant being a complete rewrite of the institution collaboration metric from a proxy-based approach to actual co-authorship counting.

1. Sankey - Show/Hide Labels Toggle

- Added a "Show Labels" checkbox in the sidebar for the Sankey diagram.
- Unchecked: flow count numbers hidden. Checked (default): numbers on flows > 1% of total.

2. Sankey - Organ Name Readability

Before:

Organ labels inside colored rectangles with small white font, nearly unreadable.

After:

Labels moved to the right side of each rectangle, white text (fontsize 8) with black outline. Always clearly readable.

3. Publication Trends - Axis Title Visibility

Before:

"Year" and "Publication Count" labels used default dark color, invisible on dark theme.

After:

Changed to light gray (#e0e0e0), clearly visible against the dark background.

4. Geo Map - Dynamic Abbreviation Legend

Before:

Short country codes (TR, TW, PL) displayed without explanation.

After:

Dynamic legend box in the top-right showing full names (e.g., TR = Turkey, TW = Taiwan). Updates with filters.

5. Institution Collaborations - Invalid Country Codes Removed

Before:

Dropdown included codes like AA, BB, BA with no known full country name.

After:

Filtered to only show countries with known full names (e.g., US = United States, CN = China).

6. Institution Collaborations - Co-Authorship Rewrite

This is the most significant change. The collaboration metric was completely rewritten to use actual co-authored publications instead of a proxy measure.

Old Method (Proxy - Shared Funder/Year)

Counted institutions sharing the same funder and year. Example: Harvard and Peking both got NIH in 2019 = +1. Problem: clustered counts (all 25-31) since many institutions share funders.

New Method (Actual Co-Authorship)

Joins three raw database tables:

- publication_author: (pmid, author_id)
- author_institution: (author_id, institution_id)
- institution: (institution_id, name, country_code)

For each paper with authors from 2+ countries, count it as a cross-country collaboration for each institution pair.

Multi-Institution Handling

Paper with Harvard (US), MIT (US), Peking (CN), Soochow (CN):

Pair	Counted?	Reason
Harvard <-> Peking	Yes (+1)	Cross-country
Harvard <-> Soochow	Yes (+1)	Cross-country
MIT <-> Peking	Yes (+1)	Cross-country
MIT <-> Soochow	Yes (+1)	Cross-country
Harvard <-> MIT	No	Same country (US)
Peking <-> Soochow	No	Same country (CN)

Data Quality Improvement

Metric	Old (Proxy)	New (Co-authorship)
File size	8.0 MB	2.0 MB
Total edges	~120,000	30,592
Count range	25-31 (clustered)	1-57 (varied)
US-China institutions	2	20 diverse
Country pairs	~200	3,785

7. Institution Collaborations - Bubble Labels

- Each institution node shows its total co-publication count in parentheses.
- Labels update dynamically with filters (Top N, country selection).
- Edge count labels placed at link origin (near source) to avoid overlap.

Files Changed

- app.py - All UI changes
- preaggregate_inst_collabs.py - New co-authorship aggregation script
- data/inst_collaborations.csv - Regenerated (2.0 MB, 30,592 rows)